Advanced Derivatives – Assignment 3

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1 Exercise 1 — Andreasen–Huge (SX5E)

1.a Data and setup

We were given SX5E implied volatilities on a strike grid from 40% to 200% of spot and several expiries. In code, we read the matrix, build absolute strikes $K = S_0 \cdot (\text{relative strike})$ with $S_0 = 100$, set $r = q = 0$, and form the maturity grid $T = \{0,\ T_1, \ldots, T_m\}$. Only cells with a quote are usable, so we keep a binary mask `mask_valid = (implied_vols > 0)` and zero out everything else. This avoids polluting the least-squares objective with missing data.

1.b From IVs to call prices

For each quoted (K_i, T_j) we turn IVs into BS call prices

$$C^{\text{mkt}}(K_i, T_j) = C_{\text{BS}}(S_0, K_i, T_j; \sigma_{i,j}, r, q)\,,$$

and set the $t = 0$ column to the intrinsic value $\max(S_0 - K, 0)$. Implementation detail: the BS routine returns the no-arbitrage limit when $\sigma \le 0$ or $T{=}t$.

1.c AH step and calibration (per expiry)

Between two consecutive expiries $T_j \to T_{j+1}$ we propagate one time step with the Andreasen–Huge fully implicit scheme [1]:

$$A_{j+1}(v)\, C_{j+1} \;=\; C_j,$$

where $A_{j+1}(v)$ is the **full** one-step matrix already including the identity. We take $\theta(K) = \text{piecewise constant}(v)$ on a partition of the strike line defined by the midpoints of the quoted indices. Let

$$z_i \;=\; \frac{\Delta t}{2\,\Delta k^2}\,\theta_{i+1}^2 \qquad (i = 1, \ldots, N{-}2),$$

then the interior rows of $A_{j+1}(v)$ are

$$A_{j+1}(v) = \text{tridiag}\big(-z_i,\ 1 + 2z_i,\ -z_i\big),$$

and the first/last rows are identity. For each j we choose v by least squares on the available strikes:

$$\min_{v} \sum_{i \in \text{quoted}(j)} \left(C^{\text{model}}_{i,j+1}(v) - C^{\text{mkt}}_{i,j+1} \right)^2.$$

Numerics: we set

$$v^{(0)} = 0.3\, S_0 \quad \text{(per parameter)}, \qquad \text{bounds } v \in [0,\, 2S_0],$$

which worked robustly. The linear system $A_{j+1}(v)\, C_{j+1} = C_j$ is solved with a single tridiagonal solve (band embedded in a sparse/dense system).

1.d Back out implied vols on the calibrated surface

Once prices $C(K_i, T_j)$ are produced on the whole grid, we invert Black–Scholes at each node to get $\widehat{\sigma}(K_i, T_j)$ by minimizing

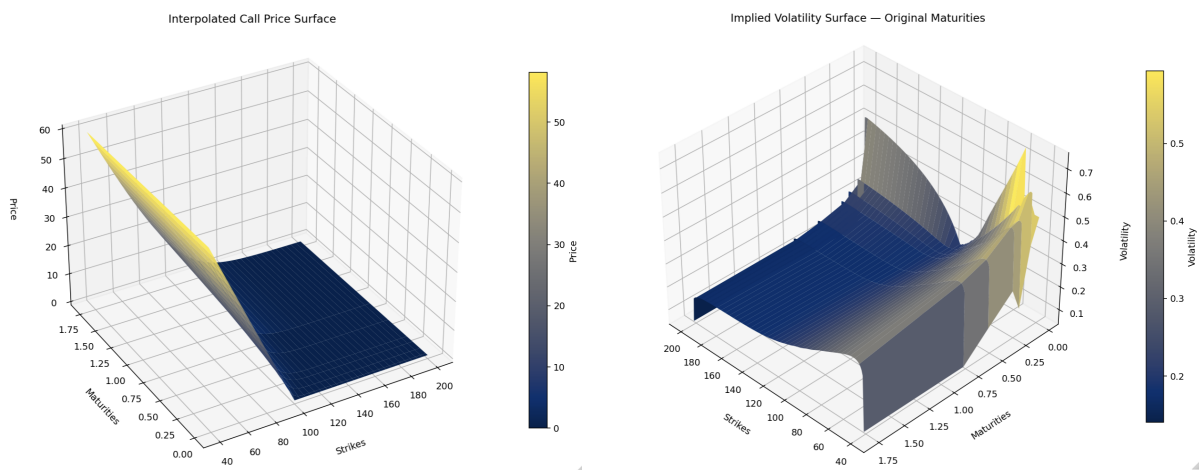
$$\min_{\sigma \geq 0} \left(C_{\text{BS}}(S_0, K_i, T_j; \sigma) - C_{i,j} \right)^2.$$

We used Nelder–Mead with seed 0.5 and record $|\hat{\sigma}|$ to guard against tiny negative steps from the unconstrained search.

1.e Extrapolation to $T = 1$ and $T = 1.5$

For a target $T_{\text{new}} \in \{1, 1.5\}$ we anchor at the previous quoted expiry $T_j < T_{\text{new}}$ and **propagate** with the same AH step using the parameters calibrated on that interval. No refit is done. We then invert BS again to obtain $\widehat{\sigma}(K, T_{\text{new}})$.

(f) Results: Surfaces and Cross-Sections

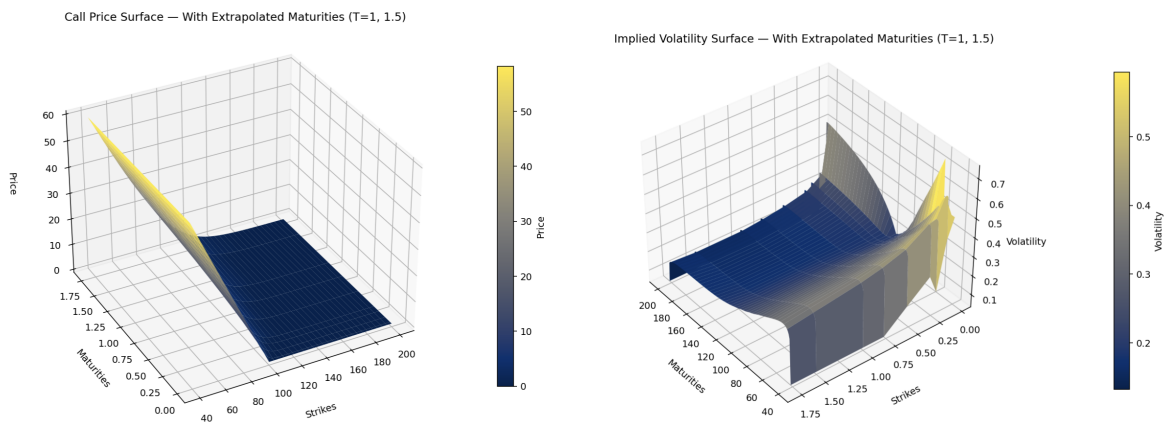


(a) Interpolated call prices on quoted maturities.

(b) Implied volatility surface on quoted maturities.

Figure A. AH interpolation on the original grid.

Comments. In **Figure A**, prices increase with maturity and decrease with strike as expected. The IV surface shows pronounced short-term wings that flatten as T grows, consistent with equity index skews.

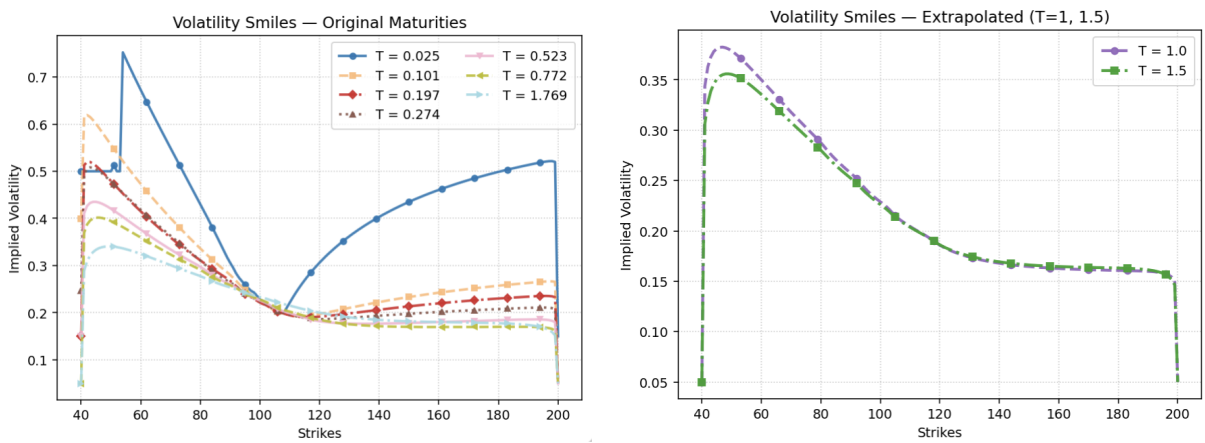


(a) Prices including $T=1$ and $T=1.5$.

(b) IV surface including $T=1$ and $T=1.5$.

Figure B. AH propagation to additional maturities.

Comments. In **Figure B**, the price surface remains monotone in T (no calendar arbitrage). The extrapolated IVs extend smoothly from the nearest quoted terms; no spurious ripples appear at the junctions.



(a) Volatility smiles at quoted maturities.

(b) Volatility smiles at $T=1$ and $T=1.5$.

Figure C. Cross-sections (strike smiles).

Comments. In **Figure C**, the shortest maturity has the largest curvature and left-skew. The extrapolated smiles lie close to their neighboring terms and preserve the overall term-structure.

Conclusion

We calibrated a local volatility surface for SX5E using the Andreasen–Huge scheme. The pipeline was: (i) read the quoted IV grid, (ii) convert to BS prices on the quoted (K, T) nodes, (iii) per maturity step, fit the piecewise–constant $\theta(K)$ by least squares and propagate with the implicit AH step, (iv) invert Black–Scholes at each node to recover $\widehat{\sigma}(K, T)$, and (v) extrapolate to $T = 1$ and $T = 1.5$ by propagating from the nearest quoted term.

The resulting price surface is monotone in maturity and decreasing in strike, as expected for calls. The implied volatility surface reproduces the observed left skew at short maturities and flattens for longer tenors. Cross-sections ('smiles') match the quoted maturities and the extrapolated curves at $T = 1$ and $T = 1.5$ sit naturally between their neighbors, without spurious oscill ations near the junctions. Overall, the AH calibration achieves a smooth interpolation in time and strike while remaining consistent with the market features visible in the raw data.

Limitations: we did not enforce hard no-arbitrage constraints beyond the implicit scheme and visual checks, and the IV inversion used an unconstrained Nelder–Mead (we guard against tiny negatives by taking the absolute value). Natural extensions would be (i) warm-starting v from the previous tenor, (ii) adding light Tikhonov regularization across K, and (iii) testing alternative seeds/optimizers to quantify sensitivity.

References

[1] Jesper Andreasen and Brian Huge, *Volatility Interpolation*, Danske Markets, March 2010. Available at `http://ssrn.com/abstract=1694972`.